ALGEBRA 1 · UNIT 9: POLYNOMIALS

Garden Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students design beds and paths for a community garden. They classify, add, subtract, and multiply polynomials to model perimeters, areas, path sizes, and costs, use special products to compare layouts, and present a garden plan of their own.

Area models make polynomial multiplication visual. Students see why $(x + 3)(x + 5)$ has four parts, why a path area is a difference of two products, and why $(x + 5)^2$ is not $x^2 + 25$. Honors students discover that a path's area depends only on the garden's perimeter.

Standards Addressed

- A-APR.1 Add, subtract, and multiply polynomials; closure
- A-SSE.1a Interpret parts of an expression
- A-SSE.2 Use structure to rewrite expressions (special products)
- A-CED.1 Use expressions in context

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: draw the garden on the board and ask how much mulch the path needs if we don't know its width. Part 1.
Day 2	Part 2: multiply polynomials with area models; path area and cost.
Day 3	Part 3: design a garden and present the plan.

Materials

- Student handout
- Graph paper or algebra tiles
- Colored pencils
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Classify, add, and subtract with guided prompts; multiply with area models; path area and one special product.
CP1	Path area by subtraction, cost polynomials, special products including $(x + a)(x - a)$, error analysis, and closure.
Honors	3-D raised beds (cubic products), a general path formula $4x^2 + Px$, fixed-perimeter comparisons, and a design with a cost constraint.

Watch For (Common Misconceptions)

- $(x + 5)^2 = x^2 + 25$ (missing the middle term).
- Forgetting to distribute the negative when subtracting polynomials.
- Adding the path width only once instead of on both sides ($12 + x$ instead of $12 + 2x$).

Extension Ideas

- Use algebra tiles to build each area.
- Get real mulch and soil prices and update the cost model.
- Partner with the school garden club.

Answer Key – CP2

Part 1: Beds & Borders

#	Task	Answer
1	Classify each polynomial by degree and number of terms: $8x + 14$; $4x^2 + 40x + 96$; 5; $x^3 - 2x$.	$8x + 14$: degree 1, linear binomial. $4x^2 + 40x + 96$: degree 2, quadratic trinomial. 5: degree 0, constant monomial. $x^3 - 2x$: degree 3, cubic binomial.
2	A bed has sides $(3x + 2)$ feet and $(x + 5)$ feet. Write and simplify its perimeter.	$2(3x + 2) + 2(x + 5) = 6x + 4 + 2x + 10 = 8x + 14$ feet
3	Two beds have areas $(x^2 + 3x)$ and $(2x^2 - x + 4)$ square feet. Find the total area.	$3x^2 + 2x + 4$ square feet
4	Fencing bed A needs $(6x + 10)$ feet and bed B needs $(4x + 2)$ feet. How much more fencing does bed A need?	$6x + 10 - 4x - 2 = 2x + 8$ feet more

Part 2: Areas & Paths

#	Task	Answer
5	An herb bed is x feet by $(x + 4)$ feet. Write its area.	$x(x + 4) = x^2 + 4x$
6	A vegetable bed is $(x + 3)$ by $(x + 5)$. Fill in the area model and write the area.	Parts: x^2, $5x$, $3x$, $15 \rightarrow x^2 + 8x + 15$
7	The main garden is 12 ft by 8 ft with a path of width x all the way around. The outside is $(12 + 2x)$ by $(8 + 2x)$. Multiply to find the total area.	$(12 + 2x)(8 + 2x) = 96 + 24x + 16x + 4x^2 = 4x^2 + 40x + 96$
8	The path's area is $4x^2 + 40x$. Find the path's area when $x = 3$ feet. Mulch costs \$1.50 per square foot. How much will the mulch cost?	$4(9) + 40(3) = 36 + 120 = 156$ sq ft; $156 \times \$1.50 = \$234$

Part 3: Design Your Garden

#	Task	Answer
9	A square flower bed has side $(x + 2)$. Write its area as a product and in standard form.	$(x + 2)^2 = x^2 + 4x + 4$
10	Draw your garden. Label each bed with binomial side lengths like $(x + 1)$ and $(x + 6)$.	Answers vary.
11	Write the total area of your beds as one polynomial. Evaluate it for $x = 2$ and find the mulch cost if the beds are covered.	Answers vary.

Answer Key – CP1

Part 1: Beds & Borders

#	Task	Answer
1	Classify $4x^2 + 40x + 96$, $x^3 - 2x$, and 7 by degree and number of terms.	Quadratic trinomial; cubic binomial; constant monomial.
2	A bed has sides $(3x + 2)$ and $(x + 5)$. Write its perimeter and its area.	Perimeter $8x + 14$; area $(3x + 2)(x + 5) = 3x^2 + 17x + 10$
3	Subtract: $(5x^2 + 3x - 2) - (2x^2 - 4x + 6)$. Then explain what the result could mean if these are the areas of two beds.	$3x^2 + 7x - 8$. It is how much larger the first bed's area is than the second's.
4	Is the sum, difference, or product of two polynomials always a polynomial? Explain, and give an example of an expression that is not a polynomial.	Yes, polynomials are closed under addition, subtraction, and multiplication. Example of a non-polynomial: $3 \div x$ or $\sqrt{x}$.

Part 2: Paths, Costs & Special Products

#	Task	Answer
5	Write the total area including the path, then subtract to find the path's area.	$(12 + 2x)(8 + 2x) = 4x^2 + 40x + 96$; path $= 4x^2 + 40x + 96 - 96 = 4x^2 + 40x$
6	Mulch costs \$1.50 per square foot. Write a polynomial for the cost of mulching the path. Find the cost when $x = 2.5$ feet.	$1.50(4x^2 + 40x) = 6x^2 + 60x$. $x = 2.5$: $6(6.25) + 60(2.5) = 37.5 + 150 = \$187.50$
7	A square bed is x by x. The board changes it to $(x + 3)$ by $(x - 3)$. Multiply and explain how the area changes.	$x^2 - 9$. The area shrinks by 9 square feet even though one side grew by 3 and the other shrank by 3.
8	Jada wrote $(x + 5)^2 = x^2 + 25$. Show that she is wrong using $x = 1$ and an area model, then write the correct product.	$x = 1$: $(1 + 5)^2 = 36$ but $1 + 25 = 26$. Correct: $x^2 + 10x + 25$ (the area model has two $5x$ rectangles).

Part 3: Design Your Garden

#	Task	Answer
9	Draw and label your beds with binomial side lengths. Include at least one square bed.	Answers vary.
10	Write polynomials for the total bed area, the outside area including a path of width x, and the path area.	Answers vary.
11	Choose x and find the total cost of mulch and fencing (fence the outside edge at \$4 per foot).	Answers vary.

Answer Key – Honors

Part 1: Raised Beds

#	Task	Answer
1	A raised bed is x ft tall, (x + 2) ft wide, and (2x − 1) ft long. Write its volume in standard form and check your answer with x = 3.	$x(x + 2)(2x − 1) = x(2x^2 + 3x − 2) = 2x^3 + 3x^2 − 2x$. Check x = 3: $3 \cdot 5 \cdot 5 = 75$ and $54 + 27 − 6 = 75$ ✓
2	Soil costs $6 per cubic foot. Write the cost polynomial and evaluate it when x = 1.5.	$6(2x^3 + 3x^2 − 2x) = 12x^3 + 18x^2 − 12x$; x = 1.5: 40.5 + 40.5 − 18 = $63
3	Multiply $(x^2 + 2x − 3)(x + 4)$. Use a table (generic rectangle).	$x^3 + 4x^2 + 2x^2 + 8x − 3x − 12 = x^3 + 6x^2 + 5x − 12$
4	Explain why the product of a degree-2 and a degree-1 polynomial has degree 3. What is the degree of the product of polynomials with degrees m and n?	The highest powers multiply: $x^2 \cdot x = x^3$. Degree m + n.

Part 2: Paths & Perimeters

#	Task	Answer
5	A garden is a ft by b ft with a path of width x around it. Show that the path's area is $4x^2 + 2(a + b)x$.	$(a + 2x)(b + 2x) − ab = ab + 2ax + 2bx + 4x^2 − ab = 4x^2 + 2(a + b)x$
6	Explain why the path area depends only on the garden's perimeter P. Rewrite the path area using P. Which two gardens need the same path mulch: 12…	$2(a + b) = P$, so path area $= 4x^2 + Px$. 12 × 8 (P = 40) and 15 × 5 (P = 40) need the same mulch; 10 × 10 also has P = 40, so all three do.
7	Several gardens have perimeter 4x. A square garden is x by x and a rectangle is (x + k) by (x − k). Show which has more area and by how much.	Square: x^2. Rectangle: $x^2 − k^2$. The square is larger by k^2.
8	Find and fix the error: $(2x − 3)^2 = 4x^2 − 9$.	$(2x − 3)^2 = 4x^2 − 12x + 9$

Part 3: Design Within a Budget

#	Task	Answer
9	Choose your garden's length and width as polynomials or numbers. Write polynomials for path area, outside perimeter, and total cost.	Answers vary.
10	Find the widest whole-number path width x that keeps the total cost within $600. Show your table of values.	Example: 12 × 8 garden: cost $= 1.50(4x^2 + 40x) + 4(40 + 8x) = 6x^2 + 92x + 160$. x = 4: 96 + 368 + 160 = $624 (too much); x = 3: 54 + 276 + 160 = $490. Widest x = 3.

11	Add raised beds and write their total soil volume as a polynomial. Present the full plan.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Original garden uses at least three beds with correct polynomial area and perimeter models. Goes beyond the minimum with a creative, realistic choice.	Original garden uses at least three beds with correct polynomial area and perimeter models.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Garden plan shows a labeled drawing, polynomial expressions for area, perimeter, and cost, and evaluated values. Polished, organized, and visually engaging.	Garden plan shows a labeled drawing, polynomial expressions for area, perimeter, and cost, and evaluated values. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.